

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0297

Flatlick 500 kV
August 27, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0297. The Interconnection Customer proposes a 936.8 MW natural gas combined cycle facility interconnecting at the Flatlick 500 kV substation in the Ohio area. The request was received on December 13, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 172 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,890,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of INTERNATIONAL PAPER - FLATLICK SUBSTATION 500 kV, the SPORN - BECK ROAD 230 kV loads to 119.7% of its 508 MVA rating. Project DFAX: 13.9%.

The MYRTLE-DATE SUBSTATION - WATERFORD POWER LLC 230 kV is overloaded to 119.2% of its 494 MVA emergency rating for the loss of the INTERNATIONAL PAPER - FLATLICK 500 kV. The project contributes 8.5% DFAX (79.3 MW) to this overload.

The GAVIN POWER LLC 500/345 kV Autotransformer is overloaded to 115.5% of its 1,310 MVA emergency rating for the loss of the INTERNATIONAL PAPER - FLATLICK 500 kV line. The project contributes 37.2% DFAX (348.2 MW) to this overload.

FISHKILL PLAINS - DILLARD-ORANGE SUBSTATION 345 kV Line: 113.1% loading (1,913 MVA) following the outage of the FLATLICK - INTERNATIONAL PAPER 500 kV; DFAX = 19.5%.

For the loss of INTERNATIONAL PAPER - FLATLICK 500 kV, the OLYMPUS 500 kV/230 kV Auto loads to 104.6% of its 1,425 MVA rating. Project DFAX: 34.8%.

OLD FORGE - PORTAGE LAKES 230 kV: 104.3% loading (694 MVA) following the outage of the WEEDPATCH - SURROWIEC 500 kV; DFAX = 7.2%.

The WEST CANTON - NORTH CAMBRIDGE 230 kV is overloaded to 103.1% of its 653 MVA emergency rating for the loss of the POINT ARENA - WEEDPATCH 500 kV. The project contributes 5.9% DFAX (55.1 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BECK ROAD - SPORN 230 kV L	Rebuild line with higher-capacity conductor	\$6,253,000	30% (\$1,879,000)
MYRTLE-DATE SUBSTATION - WATERFORD POWER LLC 230 kV	Rebuild line with higher-capacity conductor	\$48,564,000	15% (\$7,296,000)
GAVIN POWER LLC 500-345 kV Auto	Replace transformer with higher-rated unit	\$28,795,000	36% (\$10,236,000)
FISHKILL PLAINS - DILLARD-ORANGE SUBSTATION 345 kV Line	Rebuild line with higher-capacity conductor	\$120,672,000	13% (\$15,189,000)
OLYMPUS 500-230 kV Auto	Replace transformer with higher-rated unit	\$49,205,000	33% (\$16,435,000)
PORTAGE LAKES - OLD FORGE 230 kV	Reconductor line	\$10,822,000	24% (\$2,610,000)
WEST CANTON - NORTH CAMBRIDGE 230 kV	Reconductor line	\$4,829,000	26% (\$1,245,000)
	Total allocated to project		\$54,890,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.